Bihar Engineering University, Patna

B.Tech. 1st Semester Examination, 2023

Course: B.Tech. *Time: 03 Hours*
Code: 105102 *Subject: Mathematics-I (Calculus and Linear Algebra)* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin
(ii) There are NINE questions in this paper
(iii) Attempt FIVE questions in all
(iv) Question No. 1 is compulsory
(v) Symbols used (if any) have their usual meanings

Q.1 Choose the correct answer of the following (Any seven question only): [2 x 7 = 14]

(a) The value of $\int_0^1 x^3 (1-x^2)^{5/2} dx$ is

(i) $\frac{\pi}{2}$ (ii) $\frac{2}{63}$ (iii) 1 (iv) None of the above

(b) The area $\iint_R \sin(x+y)\, dxdy$ over R $\{(x,y): 0 \le x \le \pi/2, 0 \le x \le \pi/2\}$

(i) 0 (ii) 1 (iii) 2 (iv) -1

(c) The matrix $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ is

(i) Invertible but not orthogonal (ii) Invertible and orthogonal
(iii) Neither invertible nor orthogonal (iv) None of the above

(d) The sequence $\{a_n\}$, where $a_n = \sqrt[n]{n}$ is

(i) Is not convergent.
(ii) Convergent and $\lim_{n\to\infty} a_n = 0$.
(iii) Convergent and $\lim_{n\to\infty} a_n = 1$
(iv) None of the above

(e) Let $f(x) = \cos(x^2), X \in R$ then

(i) f is uniformly continuous
(ii) f is continuous, but not uniformly continuous
(iii) f continuous but unbounded
(iv) f is Lipschitz continuous

(f) $\lim_{(x,y)\to(1,2)} \frac{x^2+7y}{x+y^2}$

(i) Does not exists
(ii) Exists and the value is 2
(iii) Exists and the value is 3
(iv) None of the above

(g) If $A = 2\hat{i} + \alpha\hat{j} + \hat{k}$, $B = \hat{i} + 3\hat{j} - 8\hat{k}$. Then A and B are orthogonal if

(i) $\alpha = -2$ (ii) $\alpha = 2$ (iii) $\alpha = -1$ (iv) $\alpha = 1$

(h) Let $x + 2y + z = 0, 3x + 4y + z = 0, x - z = 0$ be a system of equations. Then

(i) It is inconsistent
(ii) It has only the trivial solution $x = 0, y = 0, z = 0$
(iii) Determinant of the matrix of coefficient is zero
(iv) None of these

(i) The value at which the Rolle's theorem is applicable for $f(x) = \cos\frac{x}{2}$ in $[\pi, 3\pi]$ is

(i) $\frac{5\pi}{2}$ (ii) $\frac{3\pi}{2}$ (iii) 2π (iv) none of the above

(j) The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$ is

(i) 3 (ii) 2 (iii) 1 (iv) none of these

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Q.2 (a) Show that : [7]

$$\int_0^{\frac{\pi}{2}} \log(\tan x + \cot x)\, dx = \pi \log 2$$

(b) Find the following limit. [7]

$$\lim_{n\to\infty} \left[t \ln\left(1 + \frac{3}{t}\right) \right]$$

Q.3 (a) Test the convergence of [8]

$$x^2 + \frac{2^2}{3.4}x^4 + \frac{2^2 4^2}{3.4.5.6.7.8}x^8 + \cdots\cdots$$

(b) Examine the convergence of the series of which the general term is [6]

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

Q.4 (a) Prove that $\log(1+x) = \frac{x}{1+\theta x}$, where $0 < \theta < 1$. Hence deduce, [8]

$\frac{x}{1+x} < \log(1+x) < x$.

(b) It is given the Rolle's theorem holds the function [6]

$f(x) = x^3 + bx^2 + cx, 1 \le x \le 2$

At the point $x = 4/3$. Find the value of b and c.

Q.5 (a) Evaluate $\int_0^{\pi/2} \frac{1}{a\cos^2 x + b\sin^2 x} dx$ [7]

(b) Discuss the convergence of the sequence whose n-*th* term is $a_n = \frac{(-1)^n}{n} + 1$. [7]

Q.6 (a) Show that the equation of the evolute of the curve [7]

$x^{2/3} + y^{2/3} = a^{2/3}$ is $(x+y)^{2/3} + (x-y)^{2/3} = 2a^{2/3}$.

(b) Find the areas of the regions that lies inside the circle $r = a\cos\theta$ and outside the cardioid $r = a(1 - \cos\theta)$. [7]

Q.7 (a) Find the solution of the following system of equations [8]

$$x_1 + 2x_2 + 2x_3 = 2$$
$$x_1 + 8x_3 + 5x_4 = -6$$
$$x_1 + x_2 + 5x_3 + 5x_4 = 3$$

Also find the basis and dimension of the solution space.

(b) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & 4 \\ 2 & 5 \end{bmatrix}$ and find its inverse. [6]

Also express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as a linear polynomial in A.

Q.8 (a) Expand $\sin x$ in Taylor's series about $x = \frac{\pi}{2}$. [6]

(b) By using Beta function evaluate $\int_0^1 x^4(1-\sqrt{x})^5 dx$. [8]

Q.9 (a) Develop the Fourier series on $-\pi < x < \pi$ to represent the function defined as [6]

$$f(x) = \begin{cases} 0, -\pi < x < 0 \\ \pi, 0 < x < \pi \end{cases}$$

(b) (i) Evaluate div $\vec{F}$ and *Curl* $\vec{F}$, where [8]

$\vec{F} = grad\,(x^2 + y^2 + z^2)e^{-\sqrt{x^2+y^2+z^2}}$.

(ii) Let $\vec{F} = (-4x - 3y + az)\hat{i} + (bx + 3y + 5z)\hat{j} + (4x + cy + 3z)\hat{k}$.
Find the values of a, b and c such that F is irrotational.

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